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College of Computer Science and Engineering

جامعة جدة

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قسم هندسة البرمجيات

CCSW 223: Software Engineering

USTS

University Smart Transport System

Integrated Software Engineering Report

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2 Introduction

University transportation plays an essential role in helping students and supervisors move between different campuses and manage daily trips efficiently. However, many of the current transportation processes are still handled manually using paper lists, WhatsApp groups, and personal estimation. These traditional methods often lead to recurring issues such as inaccurate seat counts, overcrowded buses, missed trip reminders, delays, and difficulties in tracking subscriptions or payments. As the number of students continues to grow, the limitations of the manual process become even more noticeable.

To address these challenges, our team proposes the **University Smart Transport System (USTS)**—a digital solution designed to automate and improve the entire transportation workflow. The system aims to provide essential features such as online booking, subscription management, trip reminders, attendance tracking, and electronic payment options. By introducing a centralized and user-friendly platform, USTS helps reduce human error, saves time for both students and supervisors, and ensures a more organized and reliable transportation experience.

This report presents the development of the USTS project following the principles of software engineering. It includes the problem analysis, interview findings, requirement identification, system modeling, and design diagrams such as the Context Diagram, Use Case Diagram, Sequence Diagrams, and Class Diagram. Together, these components offer a complete overview of how the system is structured and how it can enhance university transportation through a modern and efficient digital approach.

3 Problem Statement

The current university transportation system depends largely on manual processes that do not scale well and often lead to delays, inefficiency, and user dissatisfaction. Seat availability is not visible in real-time, causing overcrowding or wasted capacity. Students frequently forget to confirm their attendance due to the lack of reminders, leading to inaccurate passenger lists. Payment and subscription management are handled manually, increasing the risk of errors and miscommunication. Supervisors struggle to organize trips because bus allocation decisions rely on personal estimates rather than accurate data. These challenges highlight the need for a modern digital system that automates transportation workflows and improves communication among students, supervisors, and administrators.

4 Interview Findings

4.1 Outcome of Interviews

Outcome: The interviews revealed an urgent need for an electronic system (mobile application) to solve problems related to manual tracking, overcrowding, and trip reminders. Key requirements include: a live seat counter, notifications, electronic payment, and supervisor reports.

4.2 Interview Questions

4.2.1 *Table 1: Interview Questions (Supervisors & Users)*

This table contains a list of questions that are commonly asked during interviews with supervisors and users to gather insights into how the current transportation system works. These questions are focused on understanding how subscriptions, passenger tracking, and bus allocation are currently managed, and how an electronic system can improve these processes.

Question	
1	How do you currently track active and expired subscriptions?
2	How do you keep track of the number of passengers for each trip/building?
3	How do you decide the number of buses allocated for each trip?
4	Have you ever faced situations where the bus was overcrowded?
5	What features in an electronic system would make your work easier and save time?

4.2.2 *Table 2: Subscription and Trip Questions (Students)*

This table lists questions specifically related to students and their experience with subscriptions and trips. The goal is to understand the challenges students face with subscription duration tracking, notification reminders, and how they interact with the transportation system. These questions also explore the possibility of introducing a mobile app for easier tracking and subscription management.

Question	
1	How do you usually subscribe to transportation (weekly/monthly/daily)?
2	Do you face any problems keeping track of your subscription duration and knowing when it ends?
3	Would it make a difference for you if you received a reminder notification before the trip at a specific time?
4	Would you like to have a counter showing how many seats are left on the trip?
5	Do you prefer to pay and renew your subscription online instead of following up manually with the organizers?
6	If there was an app, would you like it to have a history of all your past trips and show how many trips you still have left in your subscription?
7	What is the biggest problem you faced with subscriptions or trips that you wish the system could solve?

4.3 **Interview Details**

4.3.1 ***Interview with Students***

Name	Job Title	Date
Leila Wail	Software Engineer	21/Sep
Bushra Hani Salman	Accounting Student	20/Sep
Maha Aqeel	Archaeology and Tourism Student	20/Sep
Janna Baitbain	Information Systems Student	20/Sep

4.3.2 ***Description of the problem (Students):***

From the interviews, the main issues faced by students include not knowing the number of available seats in each trip, forgetting to confirm attendance which leads to overcrowding or losing seats, and the absence of reminders for trip times which may cause delays or missing the bus. In addition, the manual payment process requires extra time and effort. Some students also prefer having a fixed schedule for attendance with the option to modify trips easily, as well as clear information on whether the bus goes to other universities.

4.3.3 ***Analysts comments (Students):***

Based on the students' answers, there is a real need to develop a university transportation management system through a mobile application. This application should include:

- Trip reminder notifications.
- A live counter showing available seats.
- Online subscription renewal and payment.
- A log showing past and remaining trips.
- A fixed attendance system with flexible modification options.
- Clear information on the bus destination if it serves multiple universities.

These features will reduce recurring issues (forgotten attendance, lack of seats, wasted time) and improve the experience for both students and supervisors.

4.3.4 ***Interview with Supervisors***

Name	Job Title	Date
Yahya Aqeel	Bus Supervisor	22/Sep
Ahlam	Assistant Manager	22/Sep

4.3.5 ***Description of the problem (Supervisors):***

Supervisors currently track active and expired subscriptions manually using Excel sheets or paper lists, which takes time and may cause errors. Tracking the number of passengers per trip was previously done through attendance in WhatsApp groups, and is now managed through polls. However, overcrowding may still occur when some students forget to confirm their attendance or when non-subscribed students join the trip. In addition, deciding the number of buses for each trip is mostly based on past experience and personal estimates rather than accurate data.

4.3.6 ***Analysts comments (Supervisors):***

An electronic system would make the supervisors' work easier and more accurate. The system should include:

- Automatic tracking of active and expired subscriptions.
- A live seat counter for each trip.
- Online payment and renewal options.
- Notifications for attendance and subscription status.
- Ready-made reports showing trips, seat usage, and subscription status.

This system would save time, reduce errors, and help supervisors organize trips more effectively.

5 Project Purpose from a Business Aspect

With the university's continuous expansion and the increasing number of students and supervisors using the inter-city transport service between Makkah and Jeddah, significant logistical and operational challenges have emerged in the current system. The existing system relies entirely on manual processes for managing subscriptions, confirming attendance, and handling bookings. This has led to:

- **Organizational Chaos:** Uneven and inefficient distribution of passengers across buses, with some buses being empty while others are overcrowded.
- **Waste of Time and Resources:** Students waste time in uncertain waiting, while supervisors and administrative staff spend excessive time on error-prone manual registration and follow-up tasks.
- **Poor User Experience:** Unclear trip schedules, unconfirmed seat availability, and a cumbersome payment process.

This project presents a necessary solution to transform the transport system into a centralized, intelligent digital platform. It aims to streamline operations, ensure efficiency, and radically improve the end-user experience (for students and supervisors), ultimately enhancing the university's reputation and overall satisfaction with its services.

5.1 Project Goals

Project goals have been formulated to be Specific, Measurable, Achievable, Relevant, and Time-bound:

1. **Automate Subscription Management:**
   Achieve full automation (100%) of the subscription process (issuance, renewal, and tracking of active vs. expired subscriptions) to eliminate manual errors and provide accurate digital records.
2. **Provide a Flexible Booking System:**
   Implement an electronic booking platform enabling at least 95% of registered students to reserve their seats and view real-time seat availability for each trip transparently.
3. **Reduce Latecomers and No-Shows:**
   Decrease the percentage of students missing trips due to forgetfulness by at least 70%, through an automated system sending reminder notifications (via email/app) 24 hours and 1 hour before the trip departure.
4. **Simplify the Payment Process:**
   Reduce the average payment completion time from 10 minutes (in the manual system) to a maximum of 2 minutes per student, by integrating a secure electronic payment gateway.
5. **Optimize Resource Planning:**
   Provide management with live reports and analytics on booking and attendance rates, helping them optimize bus allocation and reduce operational costs by at least 15%.

5.2 **Project Preliminary Report**

5.2.1 ***The Problem***

The current university transport management system suffers from major shortcomings at multiple levels:

- **Lack of Real-time Visibility:**
  Students have no way of knowing the number of available seats on buses before arriving at the pickup point, leading to unorganized competition and random overcrowding. Management also lacks instant data on passenger numbers to accurately plan the required number of buses.
- **Operational Inefficiency:**
  Reliance on paper records and manual processes for attendance registration and subscription management causes:
  - Significant waste of time and effort for both staff and students.
  - High rates of human error (e.g., incorrect registration, lost documents).
  - Difficulty tracking revenue and outstanding payments.
- **Poor Communication and Coordination:**
  The absence of a unified and effective communication channel to inform students of any schedule changes or remind them of their trips leads to:
  - A significant number of students missing their buses.
  - Widespread frustration and dissatisfaction with the service.
- **Cumbersome Payment Process:**
  The current system requires students to pay in cash or through other non-integrated means, increasing the time and effort required and hindering the subscription renewal process.

5.2.2 ***Findings***

Based on an analysis of the current system and interviews with a sample of users (students, supervisors, administrators), the following findings were established:

- 85% of students complain about not knowing seat availability in advance.
- Approximately 30% of students regularly forget to confirm their attendance for trips, disrupting bus allocation plans.
- 70% of students reported missing their bus at least once due to forgetting the schedule or being uncertain about it.
- The manual payment process takes an average of 10 minutes per student and consumes significant effort from administrative staff.
- Supervisors reported that student distribution across buses is random and inefficient on 40% of days.

5.2.3 *Proposed Solution*

To address the root causes of the problems mentioned above, the development and implementation of an **Integrated Digital University Transport Management System** is proposed, featuring the following:

- **Smart Booking and Notification Platform:**
  - **Digital Booking Dashboard:** Allows students to view trips, available seats, and book their seat in advance with a few clicks.
  - **Automated Notification System:** Sends booking confirmations and pre-programmed trip reminders.
- **Automated Payment and Subscription Management:**
  - **Electronic Payment Gateway:** Enables fast and secure payment via multiple channels (credit/debit cards, digital wallets).
  - **Automatic Subscription Management:** Handles automatic renewal and classification of subscriptions (active/expired).
- **Administrative Dashboard:**
  - Provides supervisors and management with live reports and analytics on:
    - Booking and attendance rates for each trip.
    - Revenue.
    - Bus utilization analysis.
  - Allows them to manage schedules and send public announcements.

5.2.4 *Cost and Schedule Estimates (Preliminary)*

- **Estimated Costs:**
  - **Development Costs (One-time):** Includes developer team fees, software/tool licenses, and UI/UX design costs.
  - **Operational & Maintenance Costs (Recurring):** Includes server/cloud hosting fees, payment gateway transaction fees, notification service costs, and costs for periodic maintenance and updates.

5.2.5 ***Proposed Project Timeline***

Phase	Duration	Key Activities & Deliverables
Phase 1: Analysis & Planning	2 Weeks	Gather detailed requirements from stakeholders (students, supervisors, admin). Model the database structure. Design UI/UX wireframes and prototypes.
Phase 2: Development & Programming	4 Weeks	Build core modules: User Management, Booking System, Payment Gateway, Admin Dashboard. Develop the front-end and back-end. Perform initial unit testing.
Phase 3: Testing & QA	2 Weeks	Conduct comprehensive system integration testing. Perform load and performance testing. Beta testing with a limited user group. Bug fixing and refinements.
Phase 4: Deployment & Training	2 Weeks	Deploy the system to the live environment. Conduct training sessions for end-users and administrators. Provide initial technical support and gather feedback.

6 Feasibility Study

6.1 Executive Summary

This feasibility study evaluates the practicality of developing a **Smart Electronic University Transport Management System** for the routes between Makkah and Jeddah.
The analysis shows that the project is both technically and financially feasible. The proposed system can be implemented using the university's existing infrastructure and modern web technologies. The expected benefits include time savings, better organization, and improved satisfaction among students and supervisors. Overall, the project is a valuable long-term investment that will enhance the efficiency of university transportation and strengthen the university's image as a digitally advanced institution. It is therefore recommended to proceed with the development of the system following an eight-week implementation plan covering analysis, development, testing, and deployment.

6.2 Problem Definition

Main Problem:

The current university transport management system between Makkah and Jeddah faces significant challenges at multiple levels, as it relies entirely on manual processes for managing subscriptions, bookings, attendance confirmation, and payments. This traditional system can no longer meet the growing needs of students and supervisors, especially with the university's continuous expansion and increasing numbers of transport service users.

Detailed Problems:

- **Lack of a Centralized Pre-booking System:**
  There is no electronic system allowing students to reserve seats in advance, leading to unorganized competition at gathering points and random overcrowding in buses.
- **Lack of Transparency in Available Seats:**
  Students cannot know the number of available seats in buses before arriving at departure points, causing general frustration and chaos in passenger distribution.
- **Poor Communication and Coordination:**
  There are no effective communication channels to inform students of schedule changes or remind them of their trip times, resulting in many students missing their trips.

- **Complexity of Manual Payment Process:**
  The current system requires students to pay in cash or through other non-integrated means, consuming considerable time and causing repeated errors in registration and accounting.

- **Inefficient Bus Distribution:**
  Management lacks real-time data on passenger numbers, hindering optimal bus allocation planning and leading to resource wastage.

- **Difficulty in Tracking Subscriptions:**
  Active and expired subscriptions are tracked through paper records and manual processes, increasing error probability and complicating periodic follow-up.

- **Negative Impact on User Experience:**
  The current system causes general frustration among students and supervisors due to inefficiency and lack of organization, negatively affecting the university's image and reputation.

6.3 Scope and Objectives of the New System

Proposed Name:

University Smart Transport System (USTS).

General Objective:

Transform the current transport system from manual to fully electronic, ensuring higher efficiency, transparency, and improved user experience (students, supervisors, and administrators).

System Description:

- Web platform and mobile application enabling students to book seats, receive reminders, and complete electronic payments.
- Supervisors' dashboard to monitor bookings, generate reports, and manage buses.
- Automated notification system to inform users of any updates.

System Distinctions:

- Integration of all processes in a single platform.
- Provision of real-time data for smarter decision-making.
- Simple and user-friendly interface.

6.4 **Market Feasibility**

The target users of this system are **University of Jeddah students** who rely on transportation between Makkah and Jeddah, along with **supervisors and administrative staff** responsible for managing these trips.

Currently, no similar electronic system exists within the university, which makes this project both necessary and innovative. Compared to external or ready-made software, the proposed system offers a customized solution that fits the university's operational environment, policies, and specific user needs. This uniqueness increases the likelihood of successful adoption after implementation.

6.5 **Alternative Solutions**

1. **Alternative 1: Full Custom System Development**
    - Includes developing a web platform, mobile application, and administrative dashboard.
2. **Alternative 2: Enhance Current System with Simple Tools**
    - Using Google Forms for booking, email notifications.
3. **Alternative 3: Adopt a Ready-made System from External Provider**
    - Purchasing a pre-prepared system and customizing it according to needs.

6.6 **Costs and Benefits of Alternatives**

Alternative	Estimated Cost	Advantages	Disadvantages
Custom System Development	20,000 - 30,000 SAR	Flexible, scalable, meets all requirements	High cost, longer development time
Current System Enhancement	5,000 - 10,000 SAR	Low cost, quick implementation	Limited functionality, not scalable
Ready-made System	20,000 - 30,000 SAR	Quick implementation, technical support	Limited customization, recurring fees

Financial Benefits of Full Development:

- 30% time savings for staff.
- 15% reduction in operational costs due to improved bus distribution.
- 40% increase in user satisfaction.

6.7 Software Impacts

- The system requires:
  - Web server to operate the platform.
  - Database (MySQL or MongoDB).
  - Integration with electronic payment gateway.
  - Support for technologies like HTML5, CSS, JavaScript, React, Node.js.
- No fundamental modifications needed in the university's current infrastructure.

6.8 Potential Organizational Changes

- Reduced reliance on manual labor for registration and follow-up processes.
- Need for technical staff to manage and maintain the system.
- Change in supervisors' roles to focus more on monitoring and analysis rather than registration.
- Enhanced university image as a developed environment keeping pace with digital transformation.

6.9 Schedule Feasibility

The project is expected to be completed within eight weeks, divided into the following main phases:

- **Analysis and Planning (2 weeks):** Collect and document system requirements.
- **Development and Programming (4 weeks):** Build, test, and refine the main system modules.
- **Testing and Deployment (2 weeks):** Perform final testing, user training, and official rollout.

This schedule is realistic and aligns with the academic timeline and available technical resources at the university.

6.10 **Risk Analysis**

Risk Type	Description	Mitigation Strategy
Development Delays	Some tasks may take longer due to technical challenges or overlapping student schedules.	Include a 10% buffer in the timeline and conduct weekly progress reviews.
Integration with Payment Gateway	Possible issues when connecting the system to external payment APIs.	Test integration early in a sandbox environment before final deployment.
Low User Adoption	Some students or supervisors might be hesitant to use a new digital system.	Provide training sessions, tutorials, and awareness campaigns before launch.

10. Recommended Alternative

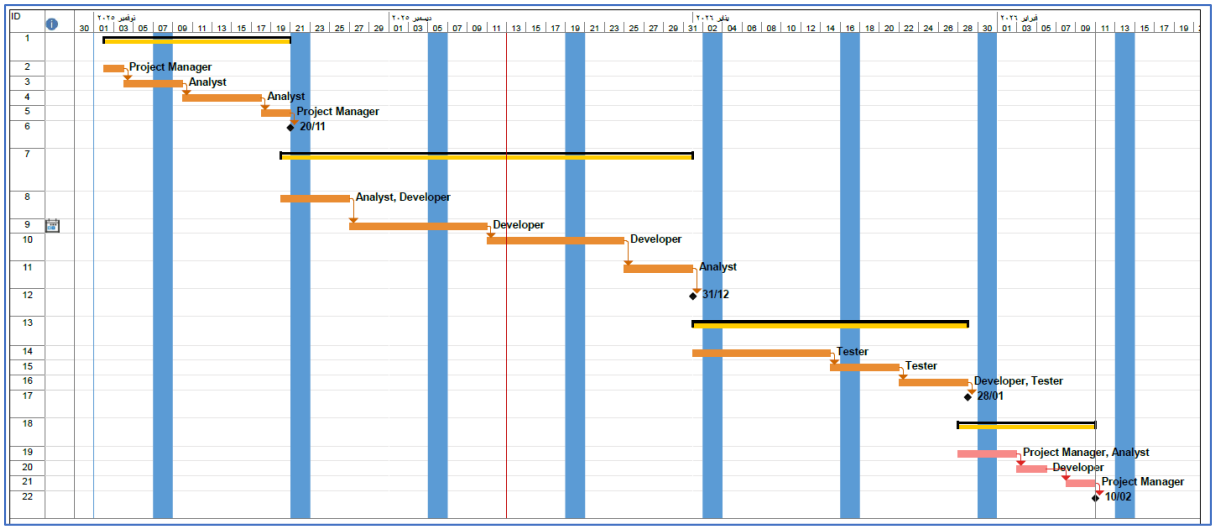
Recommended Alternative: Full Custom System Development.

Reasons:

- Meets all user needs accurately.
- Scalable for future development.
- Provides full control over data and development process.
- Despite higher cost, its long-term return is greater than other alternatives.

6.11 **Project plan**

Task Mod	Task Name	Duration	Start	Finish	Predecessors	Resource Names	Add New Column
	Phase 1: Planning and Analysis	**15 days?**	**أ ٠٢/١١/٢٥**	**خ ٢٠/١١/٢٥**			
	Select project topic	2 days	أ ٠٢/١١/٢٥	إ ٠٣/١١/٢٥		Project Manager	
	Gather requirements	4 days	ث ٠٤/١١/٢٥	أ ٠٩/١١/٢٥	٢	Analyst	
	Prepare feasibility study	6 days	إ ١٠/١١/٢٥	إ ١٧/١١/٢٥	٣	Analyst	
	Create project plan draft	3 days	ث ١٨/١١/٢٥	خ ٢٠/١١/٢٥	٤	Project Manager	
	Planning phase complete (Milestone)	0 days	خ ٢٠/١١/٢٥	خ ٢٠/١١/٢٥	٥	Project Manager	
	Phase 2: System Design and Development	**30 days**	**خ ٢٠/١١/٢٥**	**أ ٣١/١٢/٢٥**			
	Design system diagrams (UML)	5 days	خ ٢٠/١١/٢٥	أ ٢٦/١١/٢٥		Analyst, Developer	
	Build prototype	10 days	خ ٢٧/١١/٢٥	أ ١٠/١٢/٢٥	٨	Developer	
	Develop main system modules	10 days	خ ١١/١٢/٢٥	أ ٢٤/١٢/٢٥	٩	Developer	
	Review and revise design	5 days	خ ٢٥/١٢/٢٥	أ ٣١/١٢/٢٥	١٠	Analyst	
	Development phase complete (Milestone)	0 days	أ ٣١/١٢/٢٥	أ ٣١/١٢/٢٥	١١	Project Manager	
	Phase 3: Testing and Evaluation	**20 days?**	**خ ٠١/٠١/٢٦**	**أ ٢٨/٠١/٢٦**			
	Prepare test cases	10 days	خ ٠١/٠١/٢٦	أ ١٤/٠١/٢٦		Tester	
	Perform testing	5 days	خ ١٥/٠١/٢٦	أ ٢١/٠١/٢٦	١٤	Tester	
	Fix issues and retest	5 days	خ ٢٢/٠١/٢٦	أ ٢٨/٠١/٢٦	١٥	Developer, Teste	
	Testing phase complete (Milestone)	0 days	أ ٢٨/٠١/٢٦	أ ٢٨/٠١/٢٦	١٦	Project Manager	
	Phase 4: Deployment and Documentation	**10 days?**	**أ ٢٨/٠١/٢٦**	**ث ١٠/٠٢/٢٦**			
	Finalize documentation	4 days	أ ٢٨/٠١/٢٦	إ ٠٢/٠٢/٢٦		Project Manager	
	Deploy system	3 days	ث ٠٣/٠٢/٢٦	خ ٠٥/٠٢/٢٦	١٩	Developer	
	Prepare presentation	3 days	أ ٠٨/٠٢/٢٦	ث ١٠/٠٢/٢٦	٢٠	Project Manager	
	Project completion (Milestone)	0 days	ث ١٠/٠٢/٢٦	ث ١٠/٠٢/٢٦	٢١	Project Manager	



7 System Analysis

7.1 Stakeholders of the Smart University Transportation System (USTS)

7.1.1 *Primary Stakeholders*

a. Students

- Use the system to book trips, make payments, and receive notifications.
- Main users of the service.

b. Supervisors / Drivers

- Record attendance, update trip status, and ensure safe transportation.

c. University Transportation Department

- Responsible for managing schedules, allocating buses, and monitoring operations.

7.1.2 *Secondary Stakeholders*

a. Administrators

- Configure system settings, update schedules, review reports, and manage user accounts.

b. IT Support Team

- Maintains the system, applies updates, fixes technical issues, and ensures system availability.

c. Finance Department

- Manages online payment processing, billing, and financial reporting.

7.1.3 *External Stakeholders*

a. Parents (Indirect Users)

- Rely on the system to ensure their children’s safety and trip reliability.

b. External Payment Gateway

- Handles secure online payment transactions.

7.2 **Scope of Work – Smart University Transportation System (USTS)**

7.2.1 *Current Situation*

The current university transportation process is mostly manual.
Students book seats through paper forms or by contacting supervisors directly.
Payments are often handled offline, attendance is recorded manually, and there is no unified digital system to track trips, schedules, or bus status.
This leads to errors, delays, unreliable data, and low operational efficiency.

7.2.2 *Work Context*

The USTS will act as a centralized digital platform that connects students, supervisors, and administrators.
The system will support activities such as:

- Trip booking and selection
- Online payment submission
- Automated reminders and notifications
- Supervisor attendance recording
- Trip status updates
- Administrative schedule management
- Report generation and analytics dashboard

All external interactions in the system come from the stakeholders identified in the context diagram (Student, Supervisor, Administrator, Finance System).

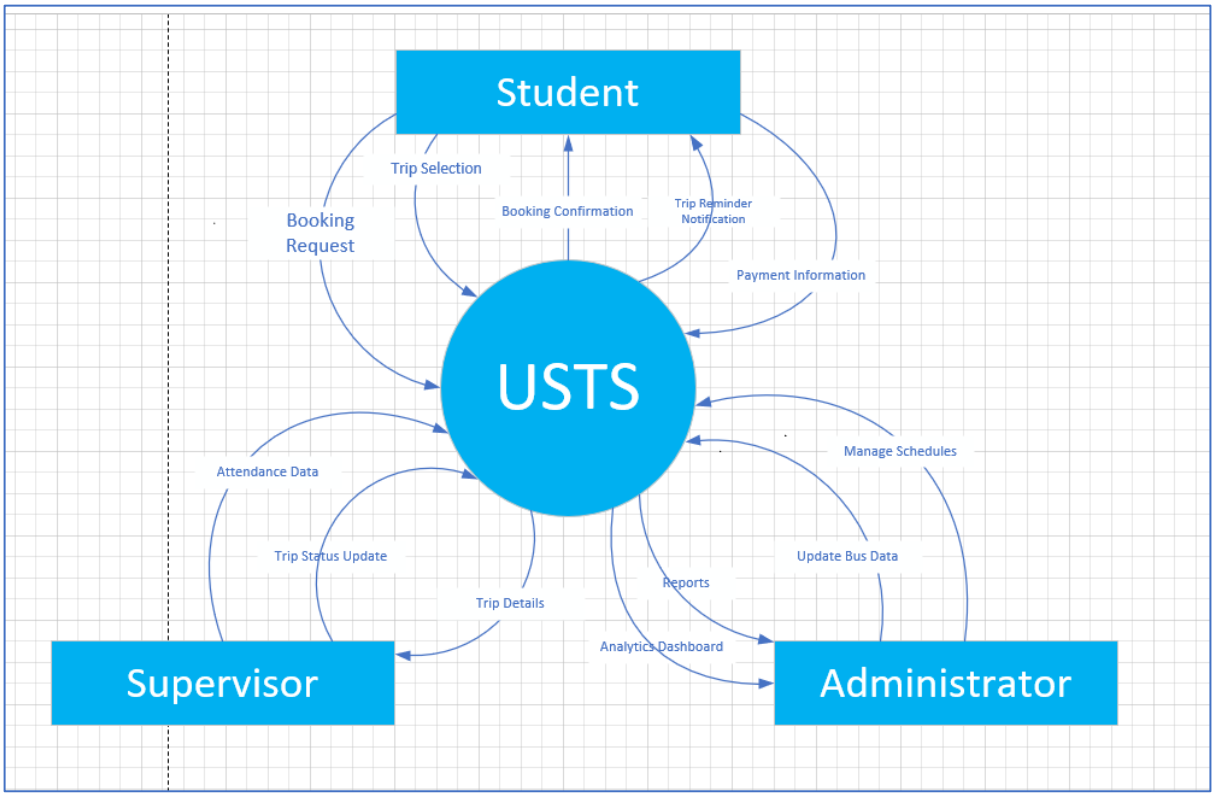
7.2.3 *Work Partitioning (Business Events)*

The work is divided into a series of business events that the system must respond to, including:

- Student submits a booking request
- System confirms the trip
- Student submits payment
- System issues a payment confirmation
- System sends trip reminders
- Supervisor records attendance
- Supervisor updates trip status
- Administrator updates schedules and bus data
- Administrator requests reports
- System generates analytics and dashboards

These events define the system boundaries and the required interactions between external entities and the USTS.

7.3 The context diagram



7.4 Event Table — Smart University Transportation System (USTS)

	Event Name	Input (from External Entity)	Output (from USTS)	Summary / Description
1	Trip Selection	Trip Selection (from Student)	—	The student selects a trip from the available schedule displayed by the system.
2	Booking Request	Booking Request (from Student)	Booking Confirmation (to Student)	The student submits a booking request, and the system confirms the reservation if seats are available.
3	Payment Submission	Payment Information (from Student)	Payment Confirmation / Receipt (to Student)	The student submits online payment for the booked trip, and the system processes and confirms the payment.
4	Trip Reminder Notification	—	Trip Reminder Notification (to Student)	The system automatically sends a reminder notification to the student before the scheduled trip time.

5	Attendance Update	Attendance Data (from Supervisor)	—	The supervisor records student attendance during the trip, and the system saves the attendance data.
6	Trip Status Update	Trip Status Update (from Supervisor)	Trip Details (to Supervisor / Administrator)	The supervisor updates the trip status such as departed, delayed, or arrived.
7	Schedule Management	Manage Schedules (from Administrator)	—	The administrator updates or modifies trip schedules within the system.
8	Bus Data Update	Update Bus Data (from Administrator)	—	The administrator updates bus-related information such as capacity, drivers, or routes.
9	Report Request	—	Reports (to Administrator)	The system generates attendance, booking, or performance reports upon administrator request.
10	Dashboard Access	—	Analytics Dashboard (to Administrator)	The system provides analytics dashboards displaying insights on trips, seat usage, and system performance.

8 System Requirements

8.1 Functional Requirements

❖ **User Registration & Authentication**

- The system shall allow students, supervisors, and administrators to register and log in using a valid university ID and password.
- The system shall verify login credentials before granting access to the system.

❖ **Subscription Management**

- The system shall allow students to view their subscription status (active, expired, remaining trips).
- The system shall allow students to renew their subscriptions electronically.

❖ **Trip Browsing & Booking**

- The system shall display available trips with seat capacity, departure time, and pickup location.
- The system shall allow students to reserve a seat on a selected trip.
- The system shall prevent students from booking a fully occupied trip.

❖ **Seat Availability Tracking**

- The system shall update the number of available seats for each trip in real time
- The system shall restrict overbooking beyond the bus capacity.

❖ **Attendance Confirmation & Tracking**

- The system shall allow supervisors to record student attendance for each trip.
- The system shall display a list of booked students for the supervisor during each trip.

❖ **Payment Processing**

- The system shall provide secure online payment to complete subscription transactions.
- The system shall generate a confirmation receipt after successful payment.

❖ **Notifications & Reminders**

- The system shall send trip reminder notifications to students before departure.

- The system shall notify students when their subscription is about to expire.

❖ **Trip Status Management**

- The system shall allow supervisors to update the status of a trip (e.g., delayed, departed, arrived).

❖ **Administrative Management**

- The system shall allow administrators to manage bus schedules, capacities, and routes.
- The system shall allow administrators to add, update, or remove buses and supervisors.

❖ **Reporting & Analytics**

- The system shall generate reports summarizing subscription data, trip attendance, and bus utilization.

8.2 Non-Functional Requirements

❖ **Usability**

- The system shall provide a simple and user-friendly interface that supports both Arabic and English.
- All main user operations shall be completed within a maximum of three clicks.

❖ **Performance**

- The system shall respond to user actions within three seconds under normal network conditions.
- The seat counter shall update instantly when a student books or cancels a trip.

❖ **Security**

- The system shall encrypt all sensitive user data including passwords and payment details.
- The system shall restrict access based on user roles (student, supervisor, administrator).

❖ **Availability & Reliability**

- The system shall be available for users 99% of the time during operational hours.

- The system shall recover from unexpected failures without data loss.

❖ **Scalability**

- The system shall support growth in the number of users and trips without affecting performance.

❖ **Maintainability**

- The system shall be designed in a modular structure to allow future updates and enhancements easily.

❖ **Compatibility**

- The system shall run smoothly on modern web browsers and mobile operating systems (Android & iOS).

9 Use Case Modeling

9.1 Use Case Diagram

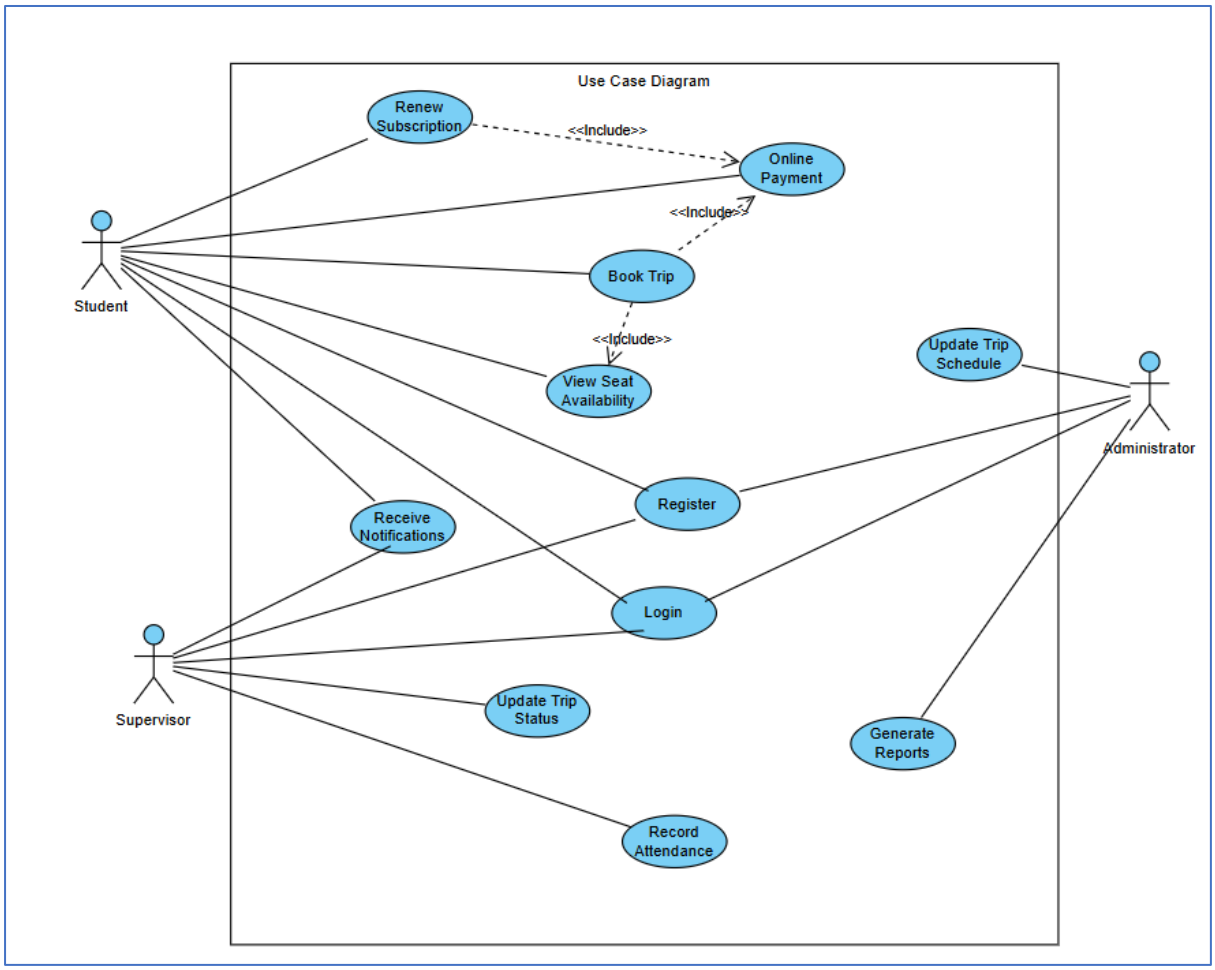


Figure 9.1: Use Case Diagram of the University Smart Transport System (USTS)

9.2 Use Case Descriptions

9.2.1 *UC1: Register*

- **Scope:** USTS
- **Primary Actor:** Student / Supervisor / Administrator
- **Precondition:** User has a valid university ID and is not registered yet.
- **Main Scenario:**
  - User opens the registration page.
  - Enters required personal and university information.
  - System validates the data.
  - System creates a new user account successfully.

9.2.2 *UC2: Login*

- **Scope:** USTS
- **Primary Actor:** Student / Supervisor / Administrator
- **Precondition:** User is already registered in the system.
- **Main Scenario:**
  - User opens the system.
  - Enters username and password.
  - System validates the credentials.
  - System grants access to the main dashboard.

9.2.3 *UC3: Renew Subscription*

- **Scope:** USTS
- **Primary Actor:** Student
- **Precondition:** Student is logged in and has an expired or expiring subscription.
- **Main Scenario:**
  - Student opens the subscription page.
  - System displays current subscription status.
  - Student selects a renewal option.
  - System proceeds to the payment process.
  - Subscription is renewed successfully.

9.2.4 *UC4: Book Trip*

- **Scope:** USTS
- **Primary Actor:** Student
- **Precondition:** Student is logged in and has an active subscription (or completes payment).
- **Main Scenario:**
  - Student views available trips.
  - System displays seat availability.
  - Student selects a trip.
  - Student confirms the booking.
  - System reserves a seat and confirms the booking.

9.2.5 *UC5: View Seat Availability*

- **Scope:** USTS
- **Primary Actor:** Student
- **Precondition:** Student is logged in.
- **Main Scenario:**
  - Student opens the trip list.
  - System shows remaining seats for each trip.
  - Student uses this information to choose a suitable trip.

9.2.6 *UC6: Online Payment*

- **Scope:** USTS
- **Primary Actor:** Student

- **Precondition:** Student initiates booking or subscription renewal that requires payment.
- **Main Scenario:**
    - Student selects the payment option.
    - System processes the payment transaction.
    - System records the payment result.
    - Student receives a payment confirmation message.

9.2.7 *UC7: Receive Notifications*

- **Scope:** USTS
- **Primary Actor:** Student
- **Precondition:** Student has booked trips or has an active subscription.
- **Main Scenario:**
    - System monitors upcoming trips and subscription status.
    - System generates reminder or status notifications.
    - Student receives notifications through the system.

9.2.8 *UC8: Record Attendance*

- **Scope:** USTS
- **Primary Actor:** Supervisor
- **Precondition:** Trip is active and students have confirmed bookings.
- **Main Scenario:**
    - Supervisor opens the attendance screen.
    - System displays the list of booked students.
    - Supervisor marks students as present or absent.
    - System saves the attendance records.

9.2.9 *UC9: Update Trip Status*

- **Scope:** USTS
- **Primary Actor:** Supervisor
- **Precondition:** Supervisor is assigned to the trip.
- **Main Scenario:**
    - Supervisor opens trip details.
    - Updates the trip status (departed, delayed, arrived).
    - System saves and updates the status for users.

9.2.10 *UC10: Update Trip Schedule*

- **Scope:** USTS
- **Primary Actor:** Administrator
- **Precondition:** Administrator is logged in.
- **Main Scenario:**
    - Administrator opens the schedule management page.
    - Adds, edits, or removes trip schedules.
    - System saves changes and updates the published schedule.

9.2.11 ***UC11: Generate Reports***

- **Scope:** USTS
- **Primary Actor:** Administrator
- **Precondition:** System contains booking, attendance, and payment data.
- **Main Scenario:**
  - Administrator selects report type and date range.
  - System collects and analyzes data.
  - System displays the generated report (with export option if needed).

10 Sequence Diagrams

10.1 Login Process Sequence Diagram

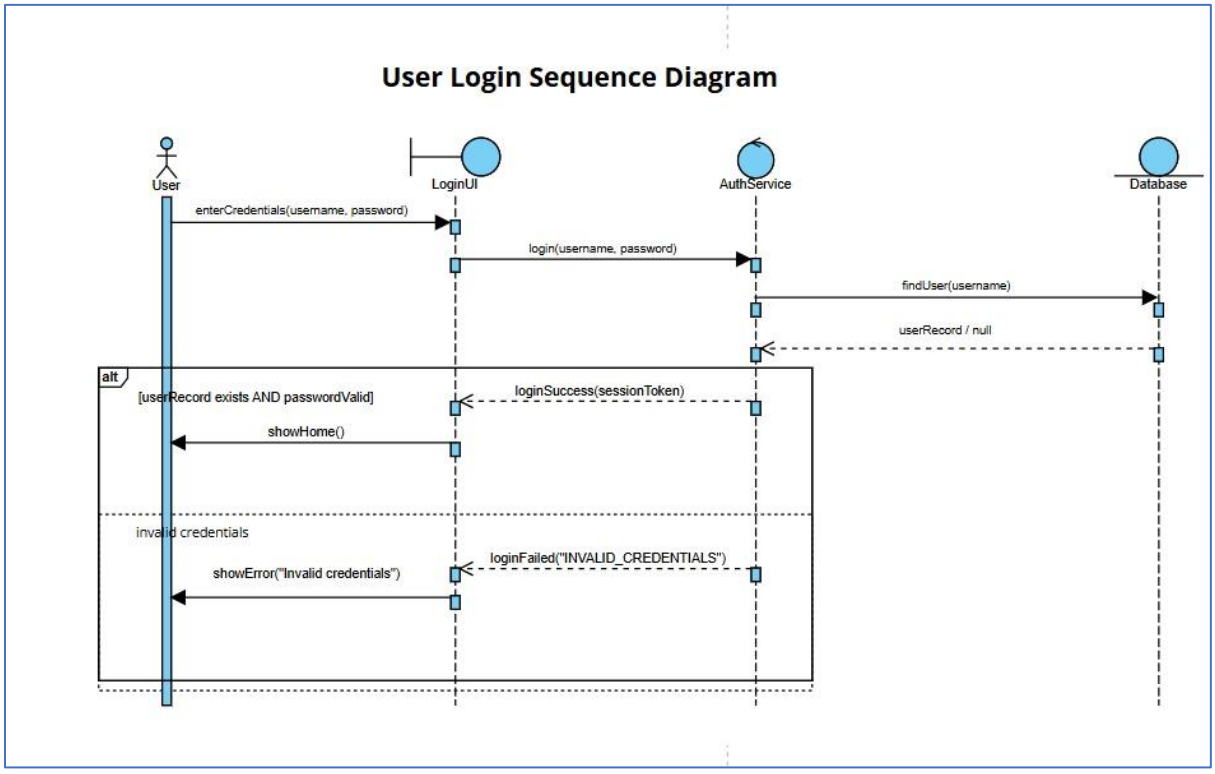


Figure 10.1: Login Process Sequence Diagram

This scenario illustrates how a user logs into the system. The process begins when the user enters their username and password through the login interface. The interface sends the credentials to the authentication service, which checks the user information in the database. If the credentials are valid, the user is successfully logged in and redirected to the home page. If the credentials are invalid, the system displays an error message indicating a failed login attempt.

10.2 **Online Payment Sequence Diagram**

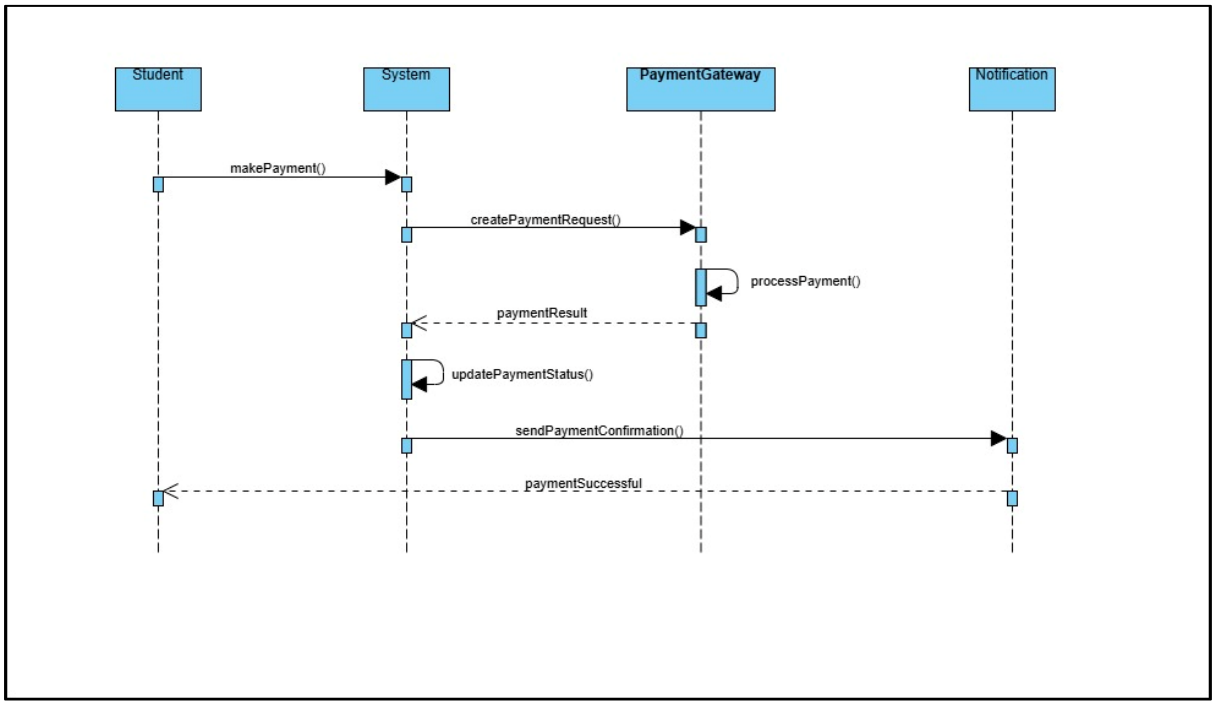


Figure 1.2Online Payment Sequence Diagram

This sequence diagram describes the online payment process within the USTS. The process starts when the student initiates a payment request. The system creates a payment request and forwards it to the payment gateway for processing. After the payment is processed, the gateway returns the payment result to the system. The system then updates the payment status and sends a confirmation notification to the student. This sequence demonstrates how secure and automated payment processing is integrated into the system.

10.3 **Supervisor Marks Student Attendance Sequence Diagram**

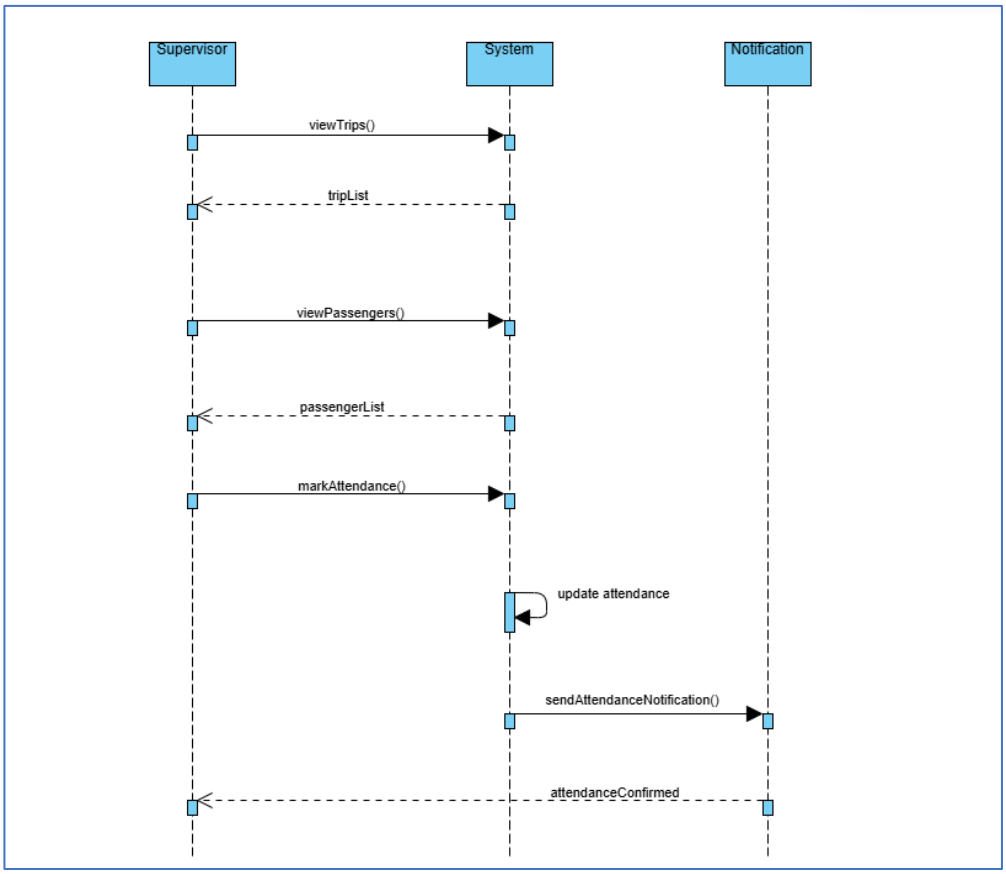


Figure 10.2: Supervisor Marks Student Attendance Sequence Diagram

This scenario explains how the supervisor manages attendance for students on a scheduled trip. It starts when the supervisor requests the list of today's trips, and the system returns the relevant information. After selecting a trip, the supervisor views the list of registered passengers. The supervisor then updates the attendance status for a student, and the system records the change. A notification is subsequently sent to inform the student of the updated attendance status. This scenario demonstrates how supervisors interact with the system to maintain accurate attendance records and improve trip organization.

10.4 **Booking a Trip Sequence Diagram**

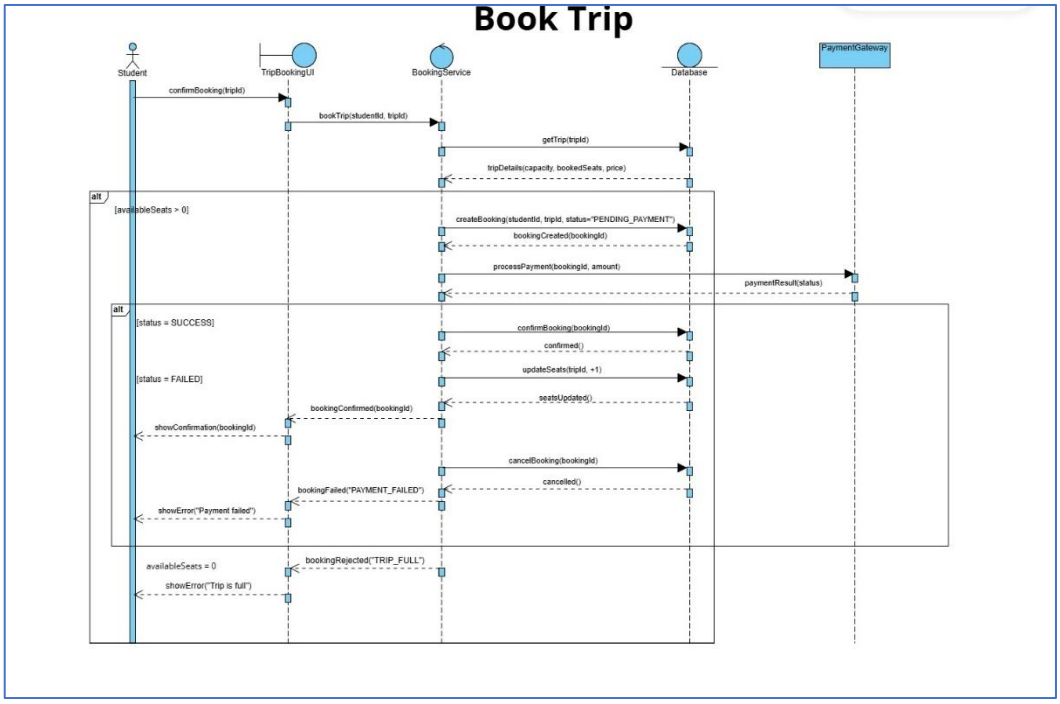


Figure 20.3: Booking a Trip Sequence Diagram

This scenario illustrates the detailed process of booking a trip in the system. The interaction starts when the student confirms the selected trip through the booking interface. The system retrieves the trip details from the database and checks the available seat capacity. If seats are available, the system creates a new booking with a pending payment status and sends a payment request to the payment gateway.

If the payment is successfully processed, the system confirms the booking, updates the trip's available seats, and displays a booking confirmation to the student. If the payment fails, the system cancels the booking and notifies the student that the payment was unsuccessful. In case no seats are available, the system rejects the booking request and informs the student that the trip is full.

This scenario demonstrates how the system handles trip reservations, seat availability checks, and payment outcomes to ensure a reliable and consistent booking process.

11 Class Diagram

11.1 Class Diagram

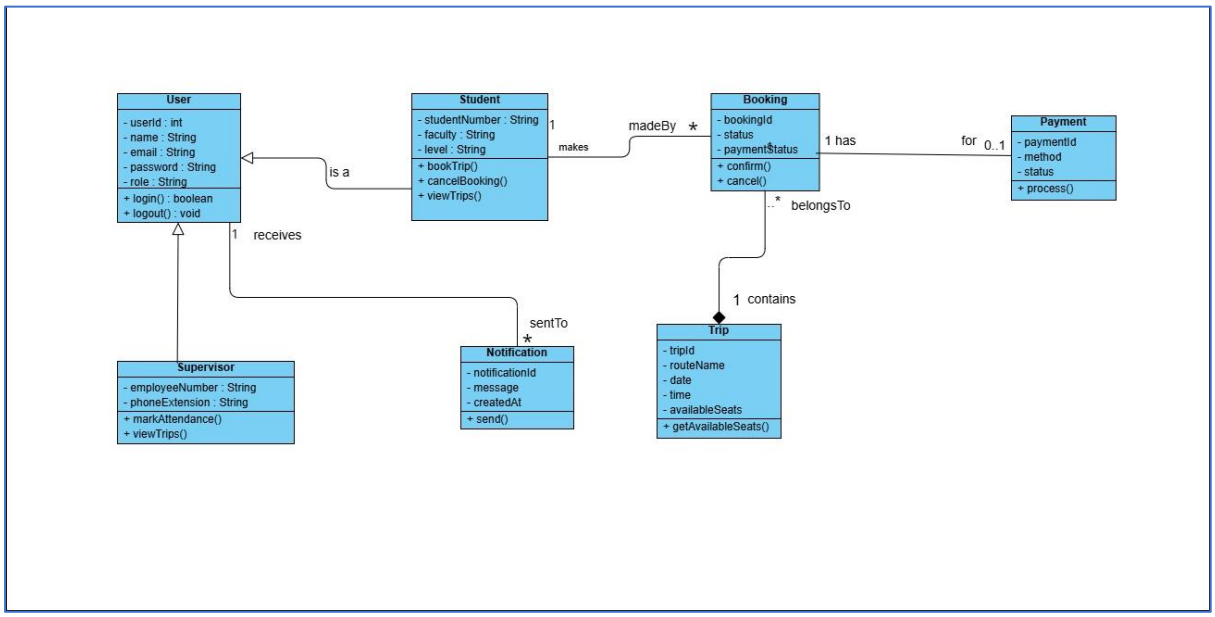


Figure 11.1: Class Diagram of the University Smart Transport System (USTS)

11.2 *Description of class diagram:*

The class diagram presents the static structure of the University Smart Transport System (USTS) after applying the final design refinements. It shows the main system classes, their attributes, operations, and relationships, while ensuring clarity and compliance with object-oriented design principles. The diagram supports the core functional requirements of the system.

11.2.1 ***User Hierarchy***

The **User** class is the base class for all system users and includes common attributes such as userId, name, email, password, and role, along with basic operations like login() and logout().
The **Student** and **Supervisor** classes inherit from User, allowing them to share a single system identifier while maintaining role-specific responsibilities.
The Student class represents students who use the transport service and provides operations such as viewing trips, booking transportation, and managing reservations.
The Supervisor class represents staff responsible for supervising trips and recording attendance, with operations focused on viewing trips and marking attendance.
This inheritance structure avoids redundancy and ensures consistent user identification across the system.

11.2.2 ***Trip and Booking Structure***

The **Trip** class models transportation trips and includes attributes such as tripId, routeName, date, time, and availableSeats.
The **Booking** class represents a student's reservation for a trip and stores information about the booking status and payment state, with operations to confirm or cancel bookings.

A **composition relationship** exists between Trip and Booking, meaning that bookings cannot exist without a related trip. Additionally, a student may have multiple bookings over time, while each booking belongs to exactly one student.

11.2.3 ***Payment and Notification Handling***

The **Payment** class manages payment transactions related to bookings, allowing each booking to have zero or one associated payment.
The **Notification** class is used to send system messages such as booking confirmations and attendance updates. Each user may receive multiple notifications, ensuring effective communication between the system and its users.

11.2.4 ***Overall Design Justification***

Overall, the updated class diagram provides a clean and consistent model of the USTS. By removing redundant identifiers, clearly defining relationships, and using inheritance and composition appropriately, the design remains simple, readable, and well-suited for academic submission while supporting all major system functionalities.

12 Conclusion

This report presented a comprehensive analysis, planning, and design of the **University Smart Transport System (USTS)** as a solution to the challenges faced by the current manual university transportation process. Through requirements analysis, stakeholder identification, feasibility study, and system modeling, the project demonstrated a clear need for a centralized and automated transportation management system.

The study showed that the proposed system is technically and financially feasible, and capable of improving operational efficiency, reducing human errors, and enhancing communication between students, supervisors, and administrators. The use of structured software engineering practices—such as use case modeling, sequence diagrams, and class diagrams—provided a clear and organized representation of the system's functionality and design.

Overall, the USTS project offers a practical and scalable digital solution that supports real-time booking, electronic payments, attendance tracking, and reporting. Implementing this system is expected to significantly improve the university transportation experience, optimize resource utilization, and contribute to a more reliable and efficient service for all stakeholders.